VibeRewrite / Local AI

Team setup and explanation guide | 23 October 2026

The existing interface now connects to a local language model through Node.js. Fixed phrase-replacement demo logic has been removed. Ollama Qwen3 4B is downloaded. Twelve real-model requests were run in two sets of six; latest outputs and errors from before final safeguards are recorded on the final page. Inspect those results before claiming semantic correctness.

How it works

Browser: HTML, CSS and JavaScript manage the input, four personas, tone controls, highlighted comparison, formats, copy and download.

Node.js: server.mjs serves the app, validates requests, sends persona and tone instructions to the selected model, and checks its response. Provider keys stay on the server.

Ollama + Qwen3 4B: Ollama runs inference on this computer. Qwen3 4B generates the improved wording and a relevant subject. No API key is needed for this local provider. Codex is the development assistant, not the runtime model.

Start on Windows

1. Start Ollama and finish downloading the model:
`ollama pull qwen3:4b`

2. Stop the old Python server if it occupies port 5173. Open PowerShell in the repository folder, not app/. With Node.js 22 or newer run:
`node --env-file-if-exists=.env server.mjs`

3. Open **http://127.0.0.1:5173/** and select Check connection. Keep Ollama and Node running. No npm install is needed. `node server.mjs` also works with local defaults. Ctrl+C stops Node.

Show the main flow

Enter a message; choose Executive, Supportive peer, Legal/compliance or Gen-Z; adjust formality, verbosity and enthusiasm; select Rewrite. Review the comparison before choosing Plain, Email, Slack or Markdown. Copy and download use that format. Email and Slack options do not send messages.

The model receives every persona and slider setting. It is instructed to preserve the language, speaker, requests, dates, amounts, uncertainty and facts while improving wording. Meaning has priority over stylistic changes.

Review and verify

Safety and limitations

Input is limited to 1,200 characters. The server limits output, concurrency and request rate, and times out at 120 seconds. Loading, Cancel, connection retry and clear error messages help explain what is happening.

Changing the input, persona, tone or format cancels a pending rewrite. A sequence guard also blocks late results. Numeric-token checks and some severity/request-reversal checks reject certain model mistakes, but they do not guarantee equivalent meaning.

A small model can misunderstand context, translate mixed-language text unexpectedly, over-edit or barely change a short sentence. CPU inference can be slow. Always review who must do what, all deadlines, amounts and negations.

Optional OpenAI later

Keep Ollama as the default. For an organizer-approved OpenAI provider, put AI_PROVIDER=openai, OPENAI_MODEL and OPENAI_API_KEY only in the ignored server .env file, then restart Node. Never put a real key in frontend code, chat or Git. OpenAI sends text to the cloud and may incur charges; local Ollama does not need a paid API key.

What has actually passed?

Passed: seven backend tests via `node --test app-tests/server.test.mjs`; JavaScript syntax; frontend state tests covering payload, stale-response prevention, cancellation, generated email subject, server errors and empty input.

Actual model: 5 of 6 requests in the second set returned accepted drafts. One mixed-language draft was rejected for changing a currency/date reference. Two accepted bodies remained verbatim. All four personas were exercised; isolated influence of every slider is not yet proven. Exact evidence follows.

Final safeguard changes: the backend now rejects unchanged bodies, newly added by tomorrow, removed explicit active, and unsupported cancellation subjects. Seven backend tests passed again. The recorded drafts predate these checks; the affected cases would now be rejected. No successful replacement for those cases has yet been established by another full model run. These partial checks do not prove semantic quality; inspect every draft.

Acceptance checklist

Try "Please send the invoice by Friday for INR 4,500". Confirm it stays a request to the recipient and preserves Friday and 4,500. Try "Kal 3 pm tak report bhej dena, budget Rs 2,500 hai" and check mixed language, time and amount. Try an unfamiliar message; compare all personas and slider extremes. Edit during a rewrite and confirm the old response never appears. Check Email subject, copy and download.

Prompt accounting and handoff

Checkpoint: 8 kit log entries; 9 known user messages; provisional ceiling 29: 9 user messages + 7 agent instructions + 13 runtime calls including the final browser check. Twelve runtime test results are recorded here; the final browser outcome is reported separately. This ceiling is not organizer-verified. The organizer decides official counting and runtime model-call treatment. Historical logs and governance files remain intact. Changes are local and uncommitted; no push or deployment was performed. The older PDF describes the earlier rule-based demo and remains as history.

Actual local-model evidence

Run: 2026-10-23T10:02:04.077Z | Model: qwen3:4b

Second set of six actual local-model calls. First-pass evidence is retained in local-model-results-first-pass.json. These status codes precede final safeguard tightening and do not prove equivalent meaning. Several drafts shown below would now be rejected. Warm second-set requests took about 0.9-4.5 seconds on this computer.

invoice request | HTTP 200

Input: Please send the invoice.

Draft: Please send the invoice.

Subject: Invoice request

ordinary bugs | HTTP 200

Input: I found a few bugs in the search filter. Can you review them tomorrow?

Draft: I've identified a few issues with the search filter. Could you please review them by tomorrow?

Subject: Search filter issues review request

deadline and amount | HTTP 200

Input: Please send the revised invoice for INR 12,500 by 25 October at 3:00 PM. Do not charge the extra INR 500.

Draft: Please send the revised invoice for INR 12,500 by 25 October at 3:00 PM. Do not charge the extra INR 500.

Subject: Revised invoice for INR 12,500 due by 25 Oct 3:00 PM

mixed language | HTTP 502

Input: Kal ki meeting 4:30 PM par shift kar do please. Budget INR 8,000 hi rakhna, extra payment approve nahi hai.

Rejected/error: The model changed a currency or date reference. Draft rejected; please retry.

casual unfamiliar | HTTP 200

Input: I cannot attend the rehearsal on Friday. Please share the recording and keep my registration active.

Draft: I'm unable to make it to Friday's rehearsal. Could you please send the recording and keep my registration active?

Subject: Friday rehearsal recording and registration update request

same meaning warmer fuller | HTTP 200

Input: I cannot attend the rehearsal on Friday. Please share the recording and keep my registration active.

Draft: I'm unable to attend the Friday rehearsal. Could you please share the recording and maintain my registration status?

Subject: Friday rehearsal recording and registration request